Accessible Power Wheelchair Storage

Ben Fisher, Camille Lu, Haley Leimbach, Will Cusato

Problem Statement

Accessible storage for those who rely on wheelchairs has long been an issue, as current solutions are either inaccessible when used independently or cumbersome, making it hard to navigate. The ability to store and carry a wide range of items is useful for every person on the planet and greatly eases certain aspects of our day. Wheelchair users wish to maintain their independence and ability to go through their daily activities without inconvenience or needing the help of others. The creation of a successful independent-use storage system for wheelchair users will greatly benefit their feeling of independence and, in turn, increase their quality of life.

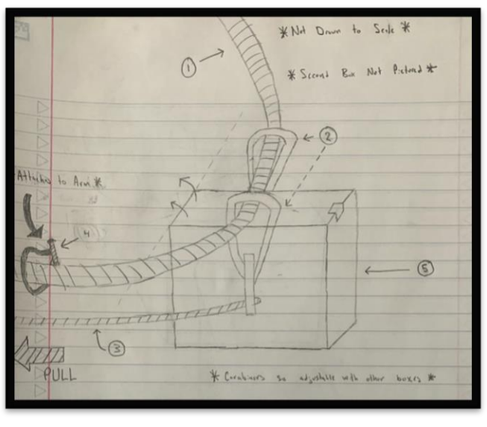
Existing Product Research

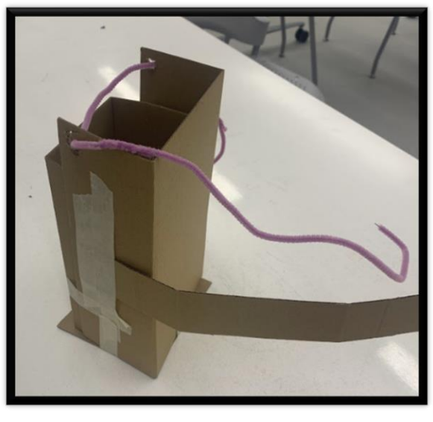
- We noticed that primary storage that is not located behind the back of the wheelchair is typically very small, without the capacity to hold many items.
- Storage located behind the wheelchair has more potential to hold items, but it's inaccessible to some users and hard to reach even if you have good mobility.
- We also saw under-chair storage that is not accessible to people with power wheelchairs as they cannot fit the storage under the chair.

User Needs & Engineering Requirements

User Need	Engineering Requirement
Accessibility	One motion needed to get storage (# steps)
Compact	Changes profile of chair by less than two inches (inches)
Independent	Reachable from within six inches of user's hand (inches)
Non-obstructive	Impedes with zero of the chair's original functions (# of obstructions)
Ease of Use	Attaches and detaches within three steps (# of steps)
Non-Destructive	Warps the frame of the chair by zero millimeters (mm)

Idea Development



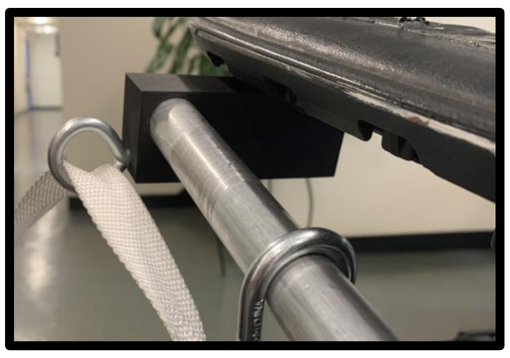


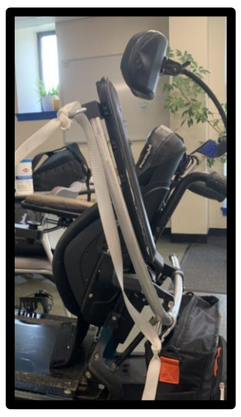
The initial concept for the product includes a flexible rail that can be morphed to better fit a variety of wheelchairs.

We decided that a rigid rail would lead to more stability. Here, it is implemented with the carabiner design.

Current Prototype

The slim nature of the aluminum track makes the profile of the chair hardly change.





The simple design and attachment of the rail allow the arm rest to be moved with the storage device attached.

Future Directions

There are a variety of changes that we would make to our device if given the opportunity to have more time. These include:

- We would have liked to make a pulley mechanism for the movement of the storage from one side to another. We believe this would make the motion of moving the storage more inclusive for all users.
- We designed specifically for the power wheelchair that was provided, and we believe that there can be an effective way to make a design that attaches to all power wheelchairs.
- Finally, we would have liked to use different-sized carabiners for the sliding-bag system. The carabiners that we ordered were barely too small. We believe slightly bigger carabiners would allow the bag to slide back when the arm is lifted due to the lack of friction.